

Sree Lekshmi Prasannan

San Francisco, California | +1(716)-253-5168 | sreelekshmip9966@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with over 4 years of experience building production AI agents, retrieval systems, statistical modeling, backend services, and big data pipelines. Shipped LLM-powered workflows, tool-calling systems, and customer-facing APIs with a focus on reliability, evaluation, and business impact.

EDUCATION

State University Of New York at Buffalo

Master of Science in Data Science

Coursework: Machine Learning, NLP, Probability, Statistical Learning, Data Mining, Time Series, Deep Learning, Probability

Buffalo, New York

Jan 2023 - Jun 2024

WORK EXPERIENCE

Profile DM Inc

Software Engineer (Artificial Intelligence Engineer)

Novi, Michigan

Jul 2025 - Present

- Launched an AI-powered recommendation microservice using a serverless LLM and RAG on AWS Lambda and FastAPI, matching users products based on profile, behavior, and catalog context.
- Orchestrated retrieval and tool-calling workflows with LangChain and LangGraph through a three-stage pipeline of vector search, cross-encoder re-ranking, and LLM personalization.
- Reduced p95 latency by 25% through prompt optimization, token trimming, and faster context retrieval with DynamoDB and Redis, improving responsiveness in production.
- Introduced model routing for low-complexity queries, cutting API costs by 70% while retaining 95% of output quality.
- Built a three-layer observability stack with input distribution tracking, statistical monitoring, LLM-as-Judge scoring, and user feedback loops to diagnose hallucination and relevance failures.
- Enforced safety guardrails following OWASP LLM and NIST AI RMF, reducing hallucinations and aligning recommendations with business and quality standards.
- Strengthened reliability with unit and integration tests for ranking, prompt templates, with CI/CD pipelines in GitHub Actions and Jenkins for automated regression checks, using LLM-as-Judge for iteration and human evaluation for final sign-off.

Community School of Arts Foundation

Software Engineer (Data Engineer)

Los Angeles, California

Jun 2024 - Jun 2025

- Developed an end-to-end sponsor big data targeting and propensity modeling workflow using PySpark, Snowflake, SQL, XGBoost, Random Forest to score and prioritize high-potential sponsors for marketing and fundraising events.
- Grew the sponsor pipeline by 40% by scraping external company data and engineering features from industry, location, engagement history, and event fit to improve prospect-to-event matching.
- Raised sponsor reply rates by 45% via model-driven outreach prioritization, and contributed to 30% more donations than the previous event cycle through data-driven sponsor selection.
- Built resilient Python ingestion workers to validate and normalize sponsor records, with versioned auditability in PostgreSQL, and used Snowflake Streams and Salesforce APIs to maintain near real-time synchronization with less than 10-minute lag.
- Delivered secure FastAPI services with JWT, RBAC, pagination, and rate limiting, then containerized and deployed them with Docker, ECR/ECS, and ALB for reliable zero-downtime rolling releases.
- Monitored ETL and service reliability via Prometheus and Grafana, reducing MTTR by 30% via alerting and operational runbooks.

University at Buffalo

Research Aide (Machine Learning Engineer)

Buffalo, New York

Oct 2023 - Jun 2024

- Trained ML models in Python/MATLAB using Random Forest, XGBoost, and K-Means on student performance and LMS data to identify weak subjects, segment cohorts, and drive targeted interventions, contributing to a 30% boost in program outcomes.
- Engineered feature pipelines from structured and unstructured data sources LMS logs, assessments, and historical grades across 8 subjects, storing structured features in MySQL for modeling, validation, and downstream analytics.
- Introduced diagnostics to flag weak areas at the subject and topic level, enabling targeted remediation strategies for students.
- Conducted ad-hoc statistical analysis of new vs. existing student performance using SQL, pandas, and A/B testing, improving intervention design and outcome tracking.
- Built Flask REST APIs, Power BI dashboards, and Streamlit apps to deliver ad-hoc analytical reports, cohort and cluster, to technical and non-technical stakeholders.

- Implemented a vendor self-service eGate pass and slot-booking system using React, Node.js/Express, and PostgreSQL, digitizing dispatch operations and supporting 400+ daily vehicle entries.
- Structured workflows for slot scheduling, shipment progress visibility, truck/driver registration, QR-based pass issuance, and gate checkpoint verification, reducing manual queue handling and improving pickup coordination.
- Connected shipment and dispatch data from internal systems and Oracle ERP via REST APIs and batch pipelines, enabling near real-time updates for order completion, pickup status, and operational dashboards.
- Containerized and deployed a CNN-based QR/code verification service with Docker, improving checkpoint validation speed by 50% and automating practical gate-entry decisions.
- Forecasted daily gate traffic via Prophet to improve slot planning and operational readiness based on historical vehicle movement patterns.
- Coordinated delivery execution through Jira-based Scrum ceremonies, including sprint planning, backlog refinement, and retrospectives, ensuring clear tracking of requirements, issues, and releases.

SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Java, MATLAB, R

AI / Agent Systems: LLM applications, AI agents, agentic workflows, RAG, tool-calling, structured outputs, LangChain, LangGraph, prompt engineering, evaluation workflows, embeddings, vector databases (Pinecone, ChromaDB)

Backend / Systems: FastAPI, Flask, REST APIs, microservices, async services, authentication/authorization (JWT, OAuth2, RBAC), caching (Redis), backend architecture

Data / Document Workflows: document ingestion, ETL/ELT, PySpark, Snowflake, Bayesian methods, big data pipelines, feature engineering, A/B testing

Cloud / Infra / Reliability: AWS (Lambda, API Gateway, S3, DynamoDB, ECS/ECR, CloudWatch), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, monitoring, Prometheus, Grafana, OpenTelemetry

Datastores: PostgreSQL, MySQL, Snowflake, DynamoDB, Redis

PROJECTS**Carbon Emission Analyzer Tech: Node.js, React, AWS Services**

- Developed an API-driven serverless backend using AWS Lambda and API Gateway for autoscaling and reduce cost
- Set up CI/CD for frontend and backend using AWS Amplify to validate and automate deployments
- Implemented a chatbot using AWS Lex to easily log purchased products and calculate carbon emission
- Integrated DynamoDB with Elasticsearch to visualize and analyze metrics in Kibana, like product vs emission

ECG Anomaly Detection with LSTM Autoencoders: PyTorch

- Built an ECG anomaly detection model in PyTorch using an LSTM autoencoder deep learning trained to learn normal heartbeat patterns from healthy ECG signals.
- Compared new ECG signals against the model's reconstructed output, using reconstruction error to identify patterns that deviated from the learned normal baseline.
- Tuned reconstruction-loss thresholds and applied preprocessing steps such as signal normalization and anomaly segmentation, improving detection sensitivity by 5%.

Reinforcement Learning with Taxi-v3

- Implemented the Q-learning algorithm from scratch in Python using OpenAI Gym's Taxi-v3 environment, building and training a tabular reinforcement learning agent without relying on high-level RL libraries.
- Designed and tuned core training components including epsilon-greedy exploration, Q-table updates, learning rate, discount factor, and exploration decay across 10,000 episodes to improve agent performance over time.
- Evaluated and visualized agent learning using episode reward tracking and environment rendering, demonstrating steady improvement from negative initial rewards to consistent successful task completion.